



The Raven surgical robot

under development for quite some time now, and as a matter of fact, Raven II is already in existence. Future plans for this surgical robot include the ability to perform heart surgery by moving the robotic arms in sync with the heartbeats, and even act autonomously without any intervention from surgeons. Sounds cool? The best part is yet to come.

Raven is a completely open source project, and the robot itself runs ROS, a flavour of Linux dedicated especially to robots. Plus, you can even get the source code of Raven by writing in to the development team.

What can it do?

To quote the development team behind Raven II, “The aim of Raven II is to develop a robot that can perform superhuman surgery, operate with dexterity and offer proper assistance to the surgeons.”

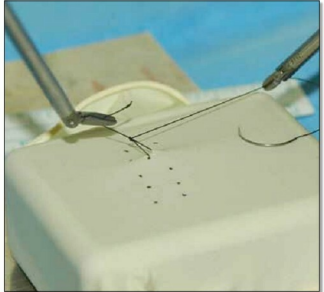
To begin with, Raven can sync its motions with the beating heart, such that when its surgical arm comes in contact with the heart tissue, it alters its movements to synchronise with the heart beats, thereby providing precision and control that no human hand can achieve by itself. In fact, when operating using Raven, the surgeon feels as if the heart is stationary and not beating—Raven does the rest.

Another function that Raven can perform is to temporarily pump or circulate blood during surgeries, in cases where the heart has stopped working. Lifesaving indeed, but this procedure is still not fool-proof, and needs a lot of work before it can be implemented on a wide scale.

Also, since nerves, muscles and bleeding can obstruct the surgeon's view at times, Raven's video camera comes in handy, as it can provide a decent 3D view of the body's tissues.

Who is involved?

Well, as already mentioned, Raven is being developed by the Universities of California and Washington. Apart from that, it also owes a good deal to Berkeley and John Hopkins University.



Raven's robotic arms tying the first knot prior to a surgical procedure (June 2006)

The project runs Robot Operating System, a version of Linux that comes with libraries and tools primarily meant for robotic applications. The OS comes with a BSD licence, and provides hardware abstraction, device drivers, libraries, and many more features. Raven is currently being tested at six different locations, and once the bugs have been fixed, we can perhaps look forward to seeing it in hospitals.

If you are interested in knowing more, visit the links given below. Alternatively, if you wish to get your hands on the source code of Raven II, simply shoot an email to the University of Washington BioRobotics Lab.



Note: All images courtesy of University of Washington BioRobotics Lab.



References

- [1] Raven Surgical Robot info page at University of Washington, BioRobotics Lab: <http://brl.ee.washington.edu/laboratory/node/26>
- [2] Raven Development Video: <http://youtu.be/Rg2xlu38PLg>
- [3] Robot Operating System: <http://www.ros.org/wiki/>
- [4] Raven Developers' Blog: <http://r2db.tumblr.com/>

By: Sufyan bin Uzayr

Sufyan bin Uzayr is a 21-year old freelance writer, graphics/Web designer, programmer and photographer. His prime areas of interest include FOSS, Web/mobile development, Web CMSs and vector art. He is also the editor of an e-journal named Brave New World (<http://www.bravenewworld.in/>). You can visit his website at <http://www.sufyan.co.nr/> or find him on Facebook at <http://facebook.com/sufyanism>.